



Food and Agriculture
Organization of the
United Nations

Future pathways for aquaculture and fisheries development in Azerbaijan

Technical Note



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Background

The Presidential Decree “On Additional Measures for the Development of Fisheries and Aquaculture in Azerbaijan”, issued on 19th December 2024, transferred the mandate for managing the aquaculture and fisheries sectors from the Ministry of Ecology and Natural Resources (MENR) to the Ministry of Agriculture (MoA). This decision reflects the government’s intention to reposition these sectors in the wider agricultural economy, and to accelerate their contribution to food security, rural development, and economic diversification. The Decree mandates the establishment of a dedicated Fisheries and Aquaculture Centre within the MoA to act as the central body responsible for guiding policy development, sector transformation and coordination.

The Food and Agriculture Organization of the United Nations (FAO) is supporting this transition through the provision of guidance on the formulation of evidence-based policies and planning frameworks, institutional strengthening and capacity development. The findings of FAO’s ongoing technical assistance, drawing on desk research, field observations, and consultations with government agencies, private sector actors and other stakeholders, has resulted in a value chain analysis and road map that proposes an initial set of recommendations to guide policy formulation, priority setting and planning processes.

Challenges

The structure of Azerbaijan’s fish supply is shaped by both limitations in capture fisheries production and untapped aquaculture potential. While aquaculture offers scope for significant production increases, the historically important capture fisheries face environmental and resource constraints that restrict opportunities for sustained growth, positioning them primarily as a tool for biodiversity conservation, stock recovery, and livelihood support amongst fishing communities.

Realizing the development potential of Azerbaijan’s fisheries and aquaculture sectors will require a coordinated reform agenda integrating governance modernisation, institutional reform, the introduction of improved production technologies and practices, research and development, training, and expanding market opportunities. Existing institutional and governance frameworks face a number of challenges. Amongst others, these include inconsistent data for decision-making, gaps in policy, legislative, and

regulatory frameworks, fragmented institutional coordination, and shortages of technical expertise and financial support mechanisms. Notably, in the aquaculture sector, the use of inefficient or outdated production technologies and management practices, combined with high input costs (feed and seed) further constrain production efficiencies and investment in the sector. Finally, low national fish consumption, limited domestic market development, and a heavy reliance on imported products collectively constrain sectoral growth, underscoring the need for coordinated strategies that stimulate demand, support import substitution, strengthen value chains, and enhance product quality and branding.

Opportunities

Global trends suggest that inland capture fisheries are increasingly constrained by ecological limitations, resource pressures and competition for water, resulting in stagnant or low levels of production. Azerbaijan is no exception to these trends, and although restocking and improved fisheries management could help rebuild stocks, the constraints suggest that capture fisheries are unlikely to become a major source of future fish production. This being the case, the primary opportunities to support capture fisheries should focus on sustainable resource management, stock recovery, and fulfilling the country’s obligations under international agreements, such as those related to the Caspian Sea Commission and the restocking of the sturgeon stocks. The restoration of culture-based fisheries through the re-establishment or optimisation of existing restocking programmes could also be used to increase the national fish supply. Additionally, attention should be paid to the social dimension by ensuring viable livelihoods are supported within fishing communities. Despite the limited potential for commercial expansion, recreational fisheries present a promising opportunity - well-managed recreational fisheries can generate significant economic benefits through tourism and related services, while also encouraging conservation, the sustainable use of aquatic resources, and providing a revenue raising instrument to support fisheries management.

In line with global and regional trends, aquaculture represents the most viable pathway for expanding Azerbaijan’s domestic fish production and can serve as the primary driver of sectoral growth by increasing the availability of fish supplies, enhancing national food security, generating employment and supporting rural livelihoods.

The way forward – the road map

The sustainable development of Azerbaijan's fisheries and aquaculture sectors requires a coherent and coordinated strategy. The following provides an outline of the road map developed by FAO. The roadmap calls for a set of strategic interventions specifically designed to enhance policy coherence, transform legal and regulatory environments to support sectoral growth, strengthen human resources and institutional capacities, improve production efficiencies and outcomes, and expand both domestic and export markets for fish products. The road map proposes focusing on three thematic areas: i) strengthening institutions and governance; ii) strengthening production and supply chains, and; iii) market development.

i. Strengthening institutions and governance

Azerbaijan's efforts to modernize its aquaculture and fisheries sector are constrained by outdated legislation, unclear institutional mandates, and fragmented governance systems that impede coherent policy formulation, sustainable resource management, and private-sector investment. Unlocking the sector's economic and environmental potential requires comprehensive institutional strengthening, anchored in a coherent national policy and supported by integrated strategies and multi-year action plans that balance economic viability, social inclusion, and environmental sustainability. There is a need for comprehensive legal and regulatory reform to improve clarity, reduce administrative burdens, introduce risk-based and spatial management tools (e.g. updated EIA processes, and the introduction of aquaculture development zones), and remove legislative and permitting barriers related to land and water use.

Establishing a comprehensive national registration system for all fish farmers and fishers forms the foundation for accurate production statistics and effective governance. Integrating registration into the Electronic Agricultural Information System (EKTIS) would significantly improve data coverage, transparency, and compliance, and could be linked to financial incentive mechanisms designed to support investment in the sector. At the same time, closer collaboration between the Fisheries and Aquaculture Centre and the State Statistical Committee is essential to shift national production statistics away from unreliable consumption-based estimates towards robust, supply-side data derived from mandatory reporting by registered producers and fishers. By embedding data collection within licensing,

permitting, and support programmes, and enforcing reporting obligations, Azerbaijan can correct current statistical distortions, and establish a credible national production database that can be used for policy formulation, sector planning, and the effective targeting of public support.

ii. Strengthening production and supply chains

A five-year **National Aquaculture Sector Development Plan** would provide a strategic instrument to guide the sustainable, inclusive, and climate-resilient growth of the aquaculture sector. The Plan should be goal-oriented and aligned with international best practice, notably the Ecosystem Approach to Aquaculture (EAA) and the FAO Guidelines for Sustainable Aquaculture (GSA). Its formulation should be informed by a comprehensive stocktaking and diagnostic assessment of the existing sector, coupled with a Strategic Environmental Assessment (SEA) that explicitly considers long-term climate change impacts. The SEA would support aquaculture zoning and the identification and designation of Aquaculture Development Zones (ADZs), ensuring that future expansion is environmentally sound, spatially planned, and socially acceptable. The plan should define a coherent strategic framework that sets out clear actions, budgets, assigned responsibilities, and implementation schedules, supported by systematic monitoring and evaluation and stakeholder consultation to ensure adaptive management and consistency with national development objectives.

Within this overarching sector planning framework, a number of tailored strategic plans that address specific sub-sectors or developmental needs should be considered. These include:

A **Sturgeon Aquaculture Strategy** would address the high economic and conservation value associated with sturgeon production. The strategy should focus on strengthening the sturgeon value chain and enhancing market competitiveness through product diversification and improved branding of Azerbaijani sturgeon products in domestic and international markets. Aquaculture is a technology-driven sector with technological development and innovation as key success factors. Amongst others, there is a need to support technology development and the optimization of production systems, and support feed development to secure feed supply chains. Strong emphasis needs to be placed on broodstock and genetic management, on-farm

biosecurity, and disease diagnostics and surveillance. The strategy should further seek to incentivize private hatchery participation in national restocking initiatives, streamline compliance with the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) and export procedures, and attract targeted investment to expand sustainable production capacity.

A **Trout Aquaculture Strategy** should prioritize improved productivity, resource use efficiencies and reduce production costs to align with international benchmarks. Amongst others, this could be achieved through the adoption of environmentally sustainable farm practices, improved farm biosecurity, siting and zoning. Developing extension services tailored to small-scale trout producers should be considered, alongside initiatives to diversify markets, strengthen product branding, and promote the development of cost-effective locally produced feeds to reduce dependency on imports.

A **Carp Aquaculture Strategy** should focus on encouraging farmers to transform their traditional extensive farming systems towards more intensive, water-efficient production systems that are integrated within broader terrestrial agricultural systems. Key priorities would include improving seed quality through upgrading hatchery infrastructure and the introduction of improved genetic strains, transitioning from supplemental feeds to compound feed use, and promoting culture-based fisheries in suitable water bodies. The strategy needs to explicitly address climate change adaptation and mitigation while encouraging the adoption of better management practices to enhance farm performance, stimulate local market development, and strengthen value chains that support small-scale rural producers.

In parallel, a **National Restocking Programme** is envisaged as a reformed and evidence-based intervention to support the regeneration of the sturgeon and Caspian salmon stocks. A technical and financial evaluation of existing restocking practices would inform revised protocols for genetic management, seed production, and release strategies. The programme should consider integrating public and private hatchery capacities under a coordinated national framework, with the dual objectives of improving stock survivorship and conserving genetic diversity in both cultured and wild populations. Long-term financial sustainability, staff capacity building, and robust monitoring systems are identified as critical success factors.

Closely linked to restocking efforts, a **National Broodstock and Genetic Resource Management Programme** should be developed to safeguard and improve genetic resources for priority aquaculture species. This programme would establish a national broodstock registry, upgrade selected government hatcheries as genetic resource centres, and introduce standardized protocols for selective breeding and cryopreservation. Specialized training, monitoring and evaluation systems, and a sustainable financing and cost-recovery plan would underpin its long-term effectiveness.

The aquaculture sector faces critical capacity constraints that limit its ability to expand production, adopt modern technologies, and develop domestic expertise in key areas such as aquatic animal health, nutrition, genetics, and technology development. A comprehensive **Capacity Development Programme** comprising a sector-wide skills needs assessment and associated implementation plan is proposed. Amongst others, the programme should be designed to strengthen aquaculture curricula at universities, expand vocational training and extension services for farmers and technicians, formally recognize industry associations, and promote industry-led research and innovation. The establishment of international partnerships should be encouraged to promote skills development and ensure technology transfers and research are closely aligned with current and future development needs.

Consideration should be given to developing a dedicated **Aquaculture Extension Services Programme** to strengthen technical knowledge and farm management skills, particularly among small-scale carp and trout farmers. Grounded in a baseline assessment of farmer needs, trained extension workers could deliver targeted training in hatchery operations, fish husbandry, biosecurity, record keeping, and farm economics. The promotion of cooperatives and producer networks could provide a useful mechanism to enhance outreach and cost-effectiveness.

A **National Aquatic Animal Health Management Programme** would address animal health and on-farm biosecurity issues across the sector - potentially structured around FAO's Progressive Management Pathway for Aquaculture Biosecurity (PMP/AB). Amongst others, the plan would strengthen diagnostic capacity at national and local laboratories, develop accredited standard operating procedures for disease diagnosis, improve on-farm biosecurity planning, disease monitoring and response, and regulate live fish movements.

A **National Aquafeed Production Programme** would aim to reduce the reliance on imported sturgeon and trout feeds while improving the quality of domestically manufactured feeds. Targeted support for investment, technology transfer and training would strengthen private sector participation and enhance the competitiveness of the national aquafeed industry.

Innovation and long-term competitiveness should further be supported by a **Technology Research and Development Programme**, that would align national research priorities to industry needs through stakeholder engagement, national R&D assessments, and the establishment of specialized university research centres. Investments in postgraduate training, public–private partnerships, and international collaboration should be coupled with robust monitoring and evaluation systems.

Finally, extending the existing **Financial Support Programme** that is supported by the Ministry of Agriculture and tailoring it to the needs of the sector offers an opportunity to stimulate sectoral growth. To avoid inefficiencies and market distortions, financial instruments must be carefully targeted, time-bound, performance-based, and inclusive. Their design needs to be sub-sector specific, and supported by strong institutional coordination and robust monitoring. To ensure fiscal sustainability, effective targeting and measurable impact, the financial support instruments should be delivered through a coherent, evidence-based subsidy framework grounded in economic realities and informed by inclusive stakeholder engagement and rigorous cost–benefit analysis.

A five-year **National Fisheries Management Plan (NFMP)** should be developed to promote the sustainable exploitation of inland and Caspian Sea fishery resources. When developing the plan, consideration should be given to the adoption of the Ecosystem Approach to Fisheries Management (EAFM). Foundational activities should include stocktaking and diagnostic analyses of the sector's fishing capacity, the socio-economic status of the fisher communities, and stock assessment surveys of all target fish resources. A review of existing management and regulatory systems should also be undertaken. The plan should establish clear

fisheries management objectives, management indicators, and monitoring systems for Caspian Sea, inland commercial, and recreational fisheries. The plan should include a sustainable finance model to manage the fisheries, integrate monitoring, control, and surveillance (MCS) systems, and align national practices with international and Caspian Sea Commission commitments.

iii. Market development

While Azerbaijan's low per capita fish consumption represents one of the most significant constraints to sectoral growth, it also provides the country with one of its greatest opportunities to support growth - even modest increases in consumption could generate substantial additional domestic demand. Unlocking this potential requires a coordinated, demand-led strategy that improves consumer awareness, affordability, product diversification and availability, integrates fish into national nutrition strategies, and invests in cold-chain and distribution systems to improve nationwide access to high-quality fish products.

While domestic carp and trout production contributes to national supply, continued reliance on imports from neighbouring countries indicates clear potential for import substitution - contingent on lowered production costs derived from improvements in production and value chain efficiencies. Achieving effective import substitution will require investment in modern production technologies and input supplies, improved farm management practices, and the expansion of processing capacity and cold-chain infrastructure. At the same time, and despite emerging growth in high-value sturgeon and caviar exports, the overall export potential for fish products remains constrained in the short- to medium-term by limited production scales, underdeveloped value chains, and strong regional competition. Building long-term international competitiveness will therefore require the development of a high-value export strategy centred on strong national branding, targeted market promotion, upgraded processing and cold-chain systems, strategic foreign investment partnerships, and improved market intelligence.

Conclusion

This preliminary roadmap provides an initial strategic orientation for the Fisheries and Aquaculture Centre, MoA, as it begins defining its mandate and priority actions to guide the development of a competitive, sustainable, and well-governed sector. While the recommendations offer a structured basis for immediate decision-making and longer-term planning, they remain contingent on further data validation, stakeholder consultation, and refinement through a participatory policy process that aligns institutional capacity with sector needs. Moving forward, there will be a need to prioritise actions based on realistic timelines, and to adopt an adaptive approach that responds to evolving technologies, markets and environmental conditions.

Acknowledgment

The authors express their sincere gratitude to Zaur Salmanli and Elshad Rzayev for their valuable contributions and support in the preparation of this technical note.

Required citation:

Shipton, T.A. & Junning, C. 2026. *Future pathways for aquaculture and fisheries development in Azerbaijan* – Technical note. Baku, FAO.

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